

SecureRoll

Organization Biometric Attendance & Verification System Project Plan Document

This document outlines the complete plan for a web-based system that verifies organization members using biometric authentication (face scan and fingerprint scan) along with ID verification, manages secure registration and login, generates digital attendance barcodes, and provides an administrative dashboard for the organization to manage members, verification, and reporting.

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1. Project Overview

SecureRoll is designed for an organization to securely verify and manage its members through a combination of biometric authentication, ID verification, and digital attendance tracking. Only registered members will be able to gain access, and the organization will have a dedicated admin panel to manage registrations, verification, and reporting.

2. Registration & Login Process

Every member of the organization must create an account before using the system. The registration form will collect the following details:

- Aadhar ID (government identity number)
- Roll Number
- Email Address
- Password (created by the user)

After account creation, the member logs in using their email and password. On successful login, the system generates a unique digital barcode for that member, which is later used for daily attendance scanning.

3. Biometric Verification (Face & Fingerprint Scan)

The website will support both face scan and fingerprint scan verification. This biometric check will only work for individuals who are already registered members of the organization — the system matches the scanned biometric data against the registered database before granting any access.

4. ID Verification

In addition to biometric checks, members must present a valid organization ID. Access is granted only if the ID is verified as authentic; if the ID is invalid or unrecognized, access is denied. This combination of biometric and ID verification adds a strong second layer of security.

5. Security & Access Control

To prevent misuse by unregistered or unauthorized individuals, the system enforces the following access control rules:

- If an unregistered person attempts to gain access more than 5 times, their mobile device is automatically frozen (blocked) from making further attempts for 24 hours.
- Full candidate/member details are visible only on the organization's authorized admin device (e.g., the admin's laptop) — restricted to admin-level access only.
- Multi-device restriction: each account can be actively logged in on only one device at a time, preventing account sharing or misuse.
- An audit log records login history, verification attempts, and barcode scan history for security monitoring.

6. Attendance System (Barcode / QR Code)

Once a member logs in, a unique barcode is generated and linked to their roll number. This barcode is presented daily and scanned to mark attendance automatically against the member's roll number. As a backup option, a QR code version will also be available, which can be scanned using a mobile camera if a dedicated barcode scanner is not available.

Optional enhancement: attendance can additionally be validated using GPS/geo-location, so that attendance is marked only when the scan happens within the organization's premises.

7. Admin / Organization Dashboard

The organization will have a dedicated admin dashboard with the following capabilities:

- View complete details of every registered member
- Support and manage the verification process for new and existing members
- Register new candidates/members into the organization's system
- View and manage daily/monthly attendance records for each member
- Approve or reject leave requests submitted by members
- Export attendance and member reports in Excel or PDF format
- Role-based access levels — e.g., Admin, Sub-admin (HR/Staff), and Member/Candidate — each with different levels of access

8. Additional Recommended Features

The following features are recommended additions to strengthen the system's security, usability, and functionality:

#	Feature	Purpose
1	OTP Verification	Confirms the person is real via email/mobile OTP during registration, before biometric/ID se
2	Forgot / Reset Password	Allows password reset using registered email or OTP verification.
3	Attendance Dashboard	Displays daily/monthly attendance history, present/absent %, and late entries per member.
4	Leave Request System	Members can apply for leave; admin can approve/reject, reflected in attendance records.
5	Notifications / Alerts	SMS or email alerts when attendance is marked or when verification fails multiple times.
6	Audit Log	Tracks login history, verification attempts, and barcode scans for accountability.
7	Role-Based Access	Separate access levels for Admin, Sub-admin/HR, and regular Members.
8	Data Encryption	Secure storage of biometric and Aadhar data using encryption best practices.
9	QR Code Backup	Alternative to barcode scanning, usable via any mobile camera.
10	Report Export	Admin can export attendance/member reports as Excel or PDF files.
11	Multi-Device Restriction	Prevents the same account from being logged in on multiple devices simultaneously.
12	Geo-Location Check	Optional GPS-based validation so attendance can only be marked within organization prem

9. Data Privacy & Security Compliance

Since the system stores highly sensitive information (biometric data and Aadhar ID numbers), it must follow strict privacy and security practices:

- All biometric and identity data must be encrypted, both in storage and in transit.
- Access to raw biometric/Aadhar data should be limited strictly to authorized admin devices.
- A clear privacy policy should be shown to members explaining how their data is used and stored.
- The system should follow applicable data protection regulations for handling biometric and government ID information.

10. Summary of Core Workflow

- 1 Member registers with Aadhar ID, Roll No, Email, and Password.
- 2 Email/mobile OTP verifies the registration.
- 3 Member logs in with email and password.
- 4 System generates a unique barcode/QR code linked to the member's roll number.
- 5 Member verifies identity using Face Scan + Fingerprint Scan + Organization ID.
- 6 If verified and registered, access is granted; the barcode is used daily for attendance.
- 7 If unregistered and access is attempted more than 5 times, the device is frozen for 24 hours.
- 8 Admin dashboard allows the organization to view member details, manage verification, register new members, track attendance, and export reports.

This plan can be extended or adjusted further as the project develops.